

PR 1

```
%{
```

```
#include<stdio.h>
```

```
%}
```

```
%%
```

```
[0-9]+ {printf("Number detected: %s\n", yytext);}
```

```
[a-zA-Z]+ {printf("Word detected: %s\n", yytext);}
```

```
%%
```

```
int yywrap() {
```

```
    return 1;
```

```
}
```

```
int main(){
```

```
    printf("enter input: \n");
```

```
    yylex();
```

```
    return 0;
```

```
}
```

PR 2

%{

#include "parser.tab.h"

#include<stdio.h>

#include<stdlib.h>

%}

%%

```
[0-9]+  {  
    yylval = atoi(yytext);  
    return NUMBER;  
}
```

```
"+"    return '+';
```

```
"-"    return '-';
```

```
"*"    return '*';
```

```
"/"    return '/';
```

```
\n     return '\n';
```

```
[ \t]  ;
```

```
.      {  
    printf("Invalid Character: %s\n", yytext);  
}
```

%%

```
int yywrap() {  
    return 1;  
}
```

```
.y
```

```
%{
```

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
int yylex();
```

```
void yyerror(char *s);
```

```
%}
```

```
%token NUMBER
```

```
%%
```

```
expr : expr '+' NUMBER
```

```
{
```

```
    printf("%d + %d = %d\n", $1, $3, $1 + $3);
```

```
    $$ = $1 + $3;
```

```
}
```

```
| expr '-' NUMBER
```

```
{
```

```
    printf("%d - %d = %d\n", $1, $3, $1 - $3);
```

```
    $$ = $1 - $3;
```

```
}
```

```
| expr '*' NUMBER
```

```
{
```

```
    printf("%d * %d = %d\n", $1, $3, $1 * $3);
```

```
    $$ = $1 * $3;
```

```
}
```

```
| expr '/' NUMBER
```

```
{
```

```
    if($3 != 0) {
```

```
        printf("%d / %d = %d\n", $1, $3, $1 / $3);
```

```
        $$ = $1 / $3;
```

```
    }
```

```
    else {
```

```
        printf("Division by zero error\n");
```

```
    }
```

```
}
```

```
| NUMBER
```

```
{
```

```
    $$ = $1;
```

```
    printf("Number: %d\n", $1);
```

```
}
```

```
;
```

```
%%
```

```
int main() {  
    printf("Enter expression: ");  
    yyparse();  
    return 0;  
}  
  
void yyerror(char *s) {  
    printf("Syntax Error: %s\n", s);  
}
```

PR 3

{

#include<stdio.h>

}

%%

"/".* { printf("It is a single line comment\n"); }

"/".**/" { printf("It is a multi line comment\n"); }

.* { printf("It is not a comment\n"); }

%%

int main()

{

printf("Enter a line:\n");

yylex();

return 0;

}

int yywrap()

{

return 1;

}

PR 4

```
%{
```

```
#include <stdio.h>
```

```
int vowels=0, letters=0;
```

```
%}
```

```
%%
```

```
[aeiouAEIOU] { vowels++; }
```

```
[a-zA-Z] { letters++; }
```

```
\n    { printf("total vowels=%d, letters=%d\n",vowels,letters); }
```

```
.    ;
```

```
%%
```

```
int yywrap() {
```

```
    return 1;
```

```
}
```

```
int main() {
```

```
    yylex();
```

```
    return 0;
```

```
}
```

PR 5

```
%{
```

```
#include<stdio.h>
```

```
int chars = 0, words = 0, spaces = 0, lines = 0;
```

```
%}
```

```
%%
```

```
[a-zA-Z0-9]+    { words++; chars += yyleng; }
```

```
[ ]           { spaces++; chars++; }
```

```
\n           { lines++; chars++; }
```

```
.            { chars++; }
```

```
%%
```

```
int main()
```

```
{
```

```
    printf("Enter text (Press Ctrl+C to stop):\n");
```

```
    yylex();
```

```
    printf("\nCharacters = %d", chars);
```

```
    printf("\nWords = %d", words);
```

```
    printf("\nSpaces = %d", spaces);
```



```
printf("\nLines = %d\n", lines);
```

```
return 0;
```

```
}
```

```
int yywrap()
```

```
{
```

```
    return 1;
```

```
}
```

PR 6

```
%{
```

```
#include<stdio.h>
```

```
int positive = 0, negative = 0;
```

```
%}
```

```
%%
```

```
-[0-9]+    { negative++; }
```

```
[0-9]+     { positive++; }
```

```
[ \t\n]    ;
```

```
.          ;
```

```
%%
```

```
int main()
```

```
{
```

```
    printf("Enter numbers(ctel+c to stop):\n");
```

```
    yylex();
```

```
    printf("\nPositive numbers = %d", positive);
```

```
    printf("\nNegative numbers = %d\n", negative);
```

```
    return 0;  
}
```

```
int yywrap()  
{  
    return 1;  
}
```

PR 7

{

#include<stdio.h>

}

%%

"int"|"float"|"char"|"if"|"else"|"while"|"for"|"return" { printf("It is a keyword\n"); }

[a-zA-Z_][a-zA-Z0-9_]* { printf("It is not a keyword\n"); }

%%

int main()

{

printf("Enter a word(ctrl+c to stop):\n");

yylex();

return 0;

}

int yywrap()

{

return 1;

}

PR 8

```
%{
```

```
#include<stdio.h>
```

```
%}
```

```
%%
```

```
[a-zA-Z_][a-zA-Z0-9_]* { printf("Valid Identifier\n"); }
```

```
.+ { printf("Invalid Identifier\n"); }
```

```
%%
```

```
int main()
```

```
{
```

```
    printf("Enter a string(ctrl + c to stop (ha saalo mai hi type karra hu)):\n");
```

```
    yylex();
```

```
    return 0;
```

```
}
```

```
int yywrap()
```

```
{
```

```
    return 1;
```

```
}
```

PR 9

{

#include<stdio.h>

}

%%

[0-9]+ { printf("Integer Constant\n"); }

[0-9]+ "." [0-9]+ { printf("Float Constant\n"); }

\[a-zA-Z]\ { printf("Character Constant\n"); }

.+ { printf("Not a Constant\n"); }

%%

int main()

{

printf("Enter a string(abhi bhi mai hi type karra hu):\n");

yylex();

return 0;

}

int yywrap()

{

return 1;

}

PR 10

```
%{
```

```
#include<stdio.h>
```

```
%}
```

```
%%
```

```
ab*      { printf("String matches the pattern(a followed by number of b's(ha mai hi type  
karra hu))\n"); }
```

```
.*       { printf("String does not match\n"); }
```

```
%%
```

```
int main()
```

```
{
```

```
    printf("Enter a string:\n");
```

```
    yylex();
```

```
    return 0;
```

```
}
```

```
int yywrap()
```

```
{
```

```
    return 1;
```

```
}
```

